In [39]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.patches as mpatches
```

Data Findings

In [9]:
```
df = pd.read_csv("bank-additional-full.csv", delimiter=";")
```

In [10]:
```
df
```

Out[10]:

	age	job	marital	education	default	housing	loan	contact	month	day_of_week	...	campai
0	56	housemaid	married	basic.4y	no	no	no	telephone	may	mon	...	
1	57	services	married	high.school	unknown	no	no	telephone	may	mon	...	
2	37	services	married	high.school	no	yes	no	telephone	may	mon	...	
3	40	admin.	married	basic.6y	no	no	no	telephone	may	mon	...	
4	56	services	married	high.school	no	no	yes	telephone	may	mon	...	
...	...	...	...	...	...	...	...	...	...	...	...	
41183	73	retired	married	professional.course	no	yes	no	cellular	nov	fri	...	
41184	46	blue-collar	married	professional.course	no	no	no	cellular	nov	fri	...	
41185	56	retired	married	university.degree	no	yes	no	cellular	nov	fri	...	
41186	44	technician	married	professional.course	no	no	no	cellular	nov	fri	...	
41187	74	retired	married	professional.course	no	yes	no	cellular	nov	fri	...	

41188 rows × 21 columns

In [11]:
```
df['y'].value_counts()
```

Out[11]:
```
no     36548
yes     4640
Name: y, dtype: int64
```

In [12]:
```
df.shape
```

Out[12]:
```
(41188, 21)
```

In [13]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   age             41188 non-null  int64  
 1   job             41188 non-null  object 
 2   marital         41188 non-null  object 
 3   education       41188 non-null  object 
 4   default         41188 non-null  object 
 5   housing         41188 non-null  object 
 6   loan            41188 non-null  object 
 7   contact         41188 non-null  object 
 8   month           41188 non-null  object 
 9   day_of_week     41188 non-null  object 
 10  duration        41188 non-null  int64  
 11  campaign        41188 non-null  int64  
 12  pdays           41188 non-null  int64  
 13  previous        41188 non-null  int64  
 14  poutcome        41188 non-null  object 
 15  emp.var.rate    41188 non-null  float64
 16  cons.price.idx  41188 non-null  float64
 17  cons.conf.idx   41188 non-null  float64
 18  euribor3m       41188 non-null  float64
 19  nr.employed     41188 non-null  float64
 20  y               41188 non-null  object 
dtypes: float64(5), int64(5), object(11)
memory usage: 6.6+ MB
```

In [14]:
```
# all input variables
inputX = [x for x in df.columns if x != 'y']
```

```
inputX
```

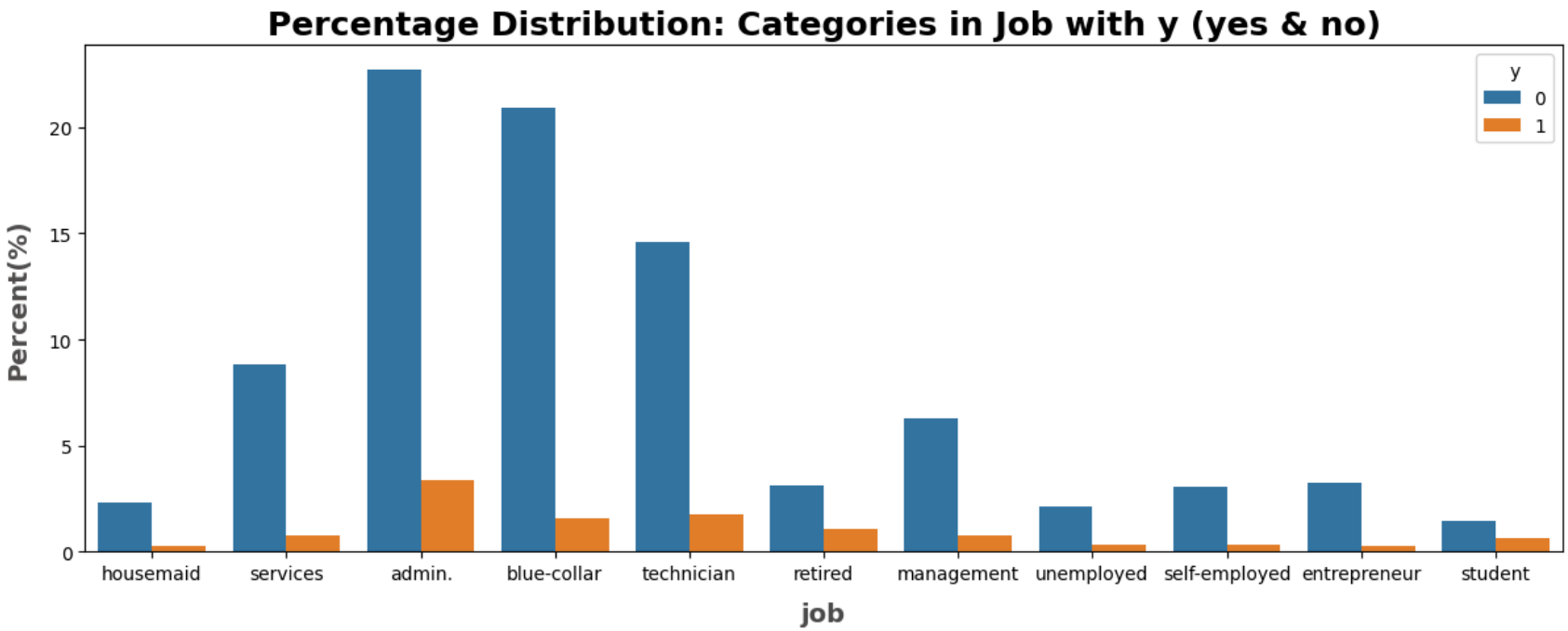
Out[14]:
```
['age',
 'job',
 'marital',
 'education',
 'default',
 'housing',
 'loan',
 'contact',
 'month',
 'day_of_week',
 'duration',
 'campaign',
 'pdays',
 'previous',
 'poutcome',
 'emp.var.rate',
 'cons.price.idx',
 'cons.conf.idx',
 'euribor3m',
 'nr.employed']
```

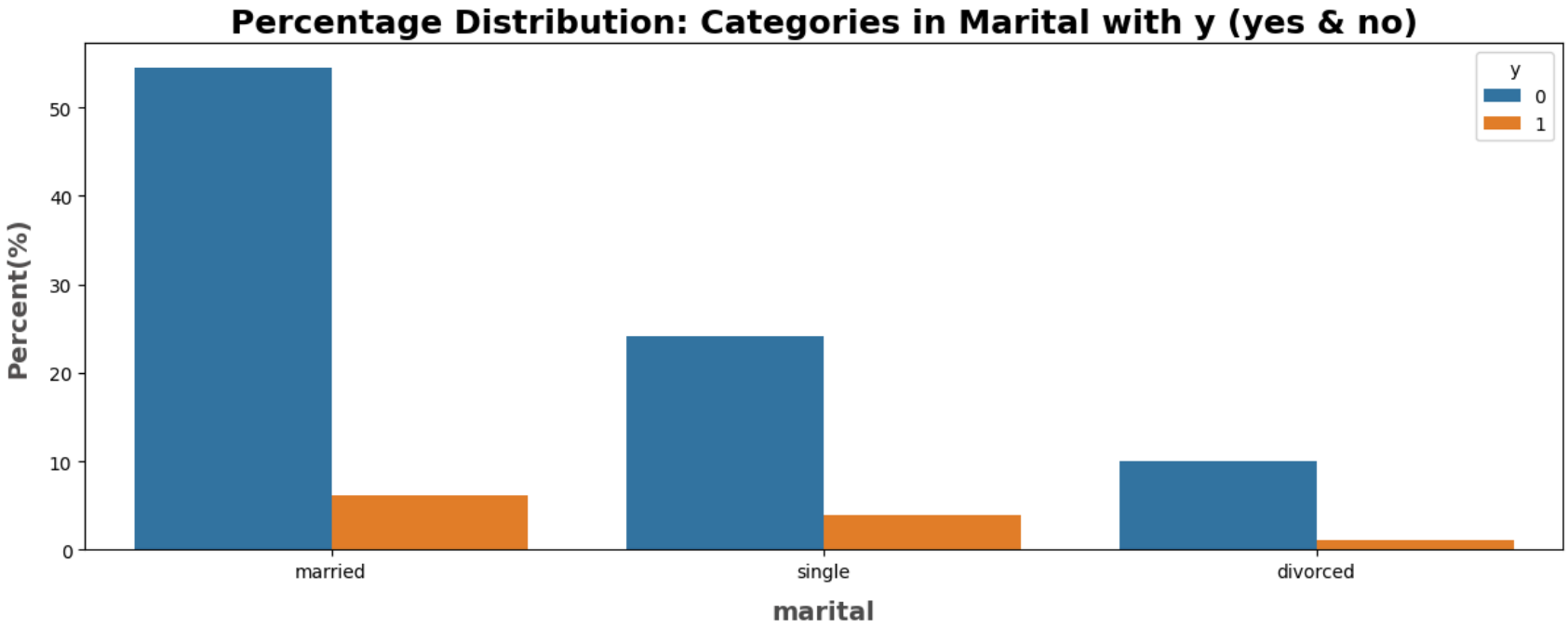
In [17]:
```
# segregate categorical and numerical variables
numericalVariables = df._get_numeric_data().columns
numericalVariables = list(set(numericalVariables))
print("Numerical Variables: ")
print(numericalVariables)
print()

categoricalVariables = list(set(inputX) - set(numericalVariables))
print("Categorical Variables: ")
print(categoricalVariables)
```

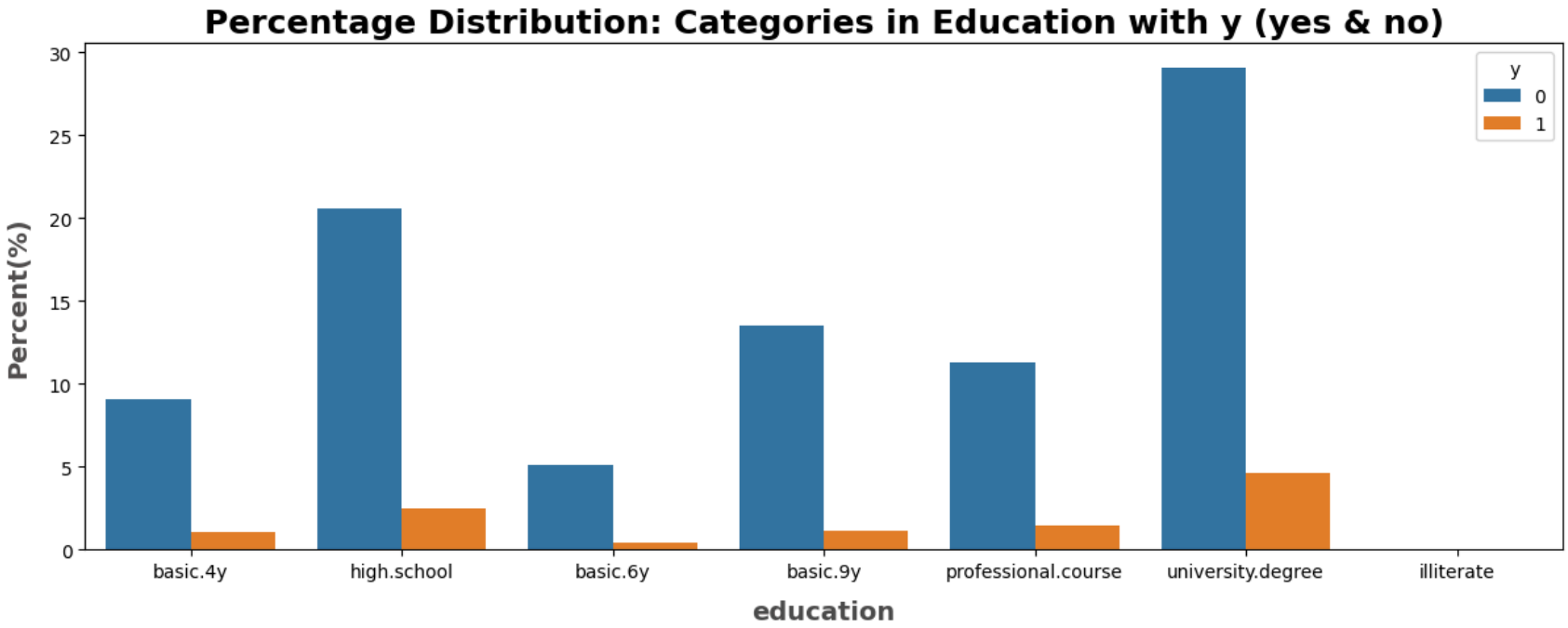
```
Numerical Variables:
['cons.conf.idx', 'emp.var.rate', 'age', 'campaign', 'previous', 'euribor3m', 'duration', 'cons.price.idx',
'nr.employed', 'pdays']

Categorical Variables:
['loan', 'job', 'contact', 'day_of_week', 'education', 'month', 'marital', 'default', 'housing', 'poutcom
e']
```

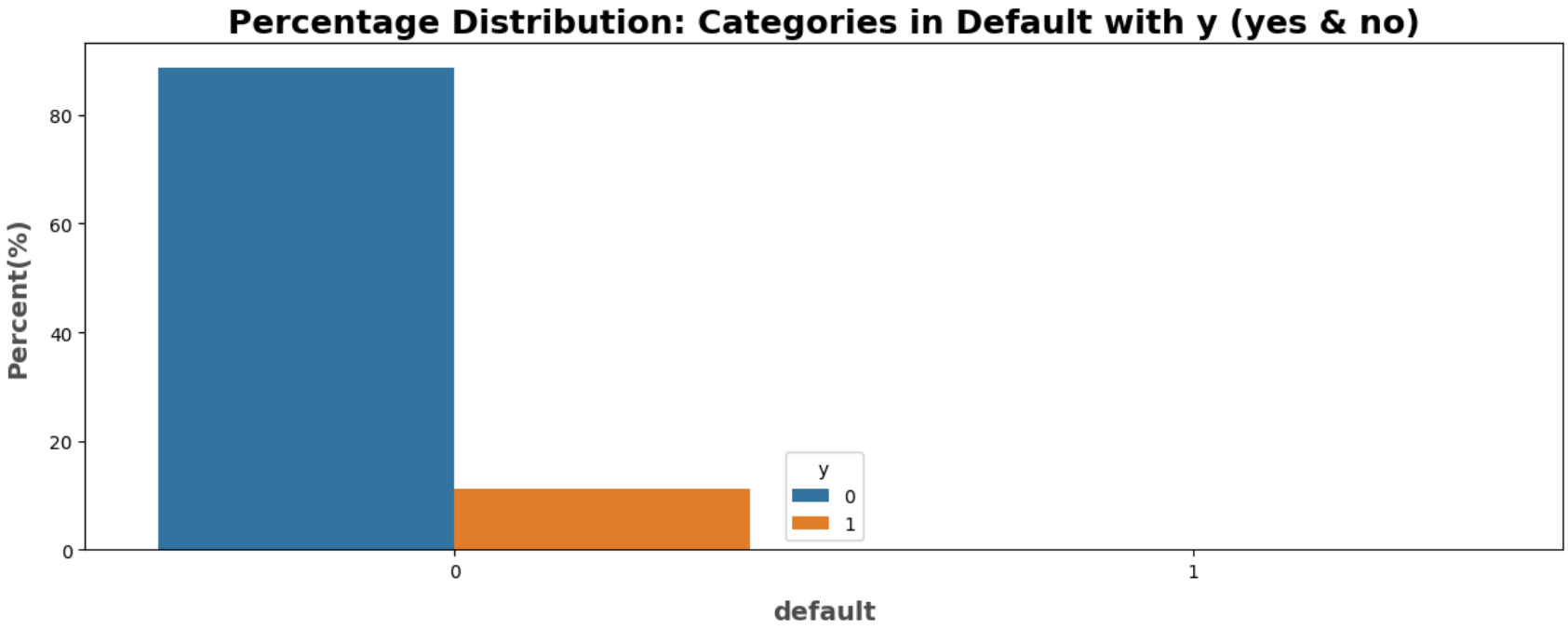
Percentage Distribution of Data

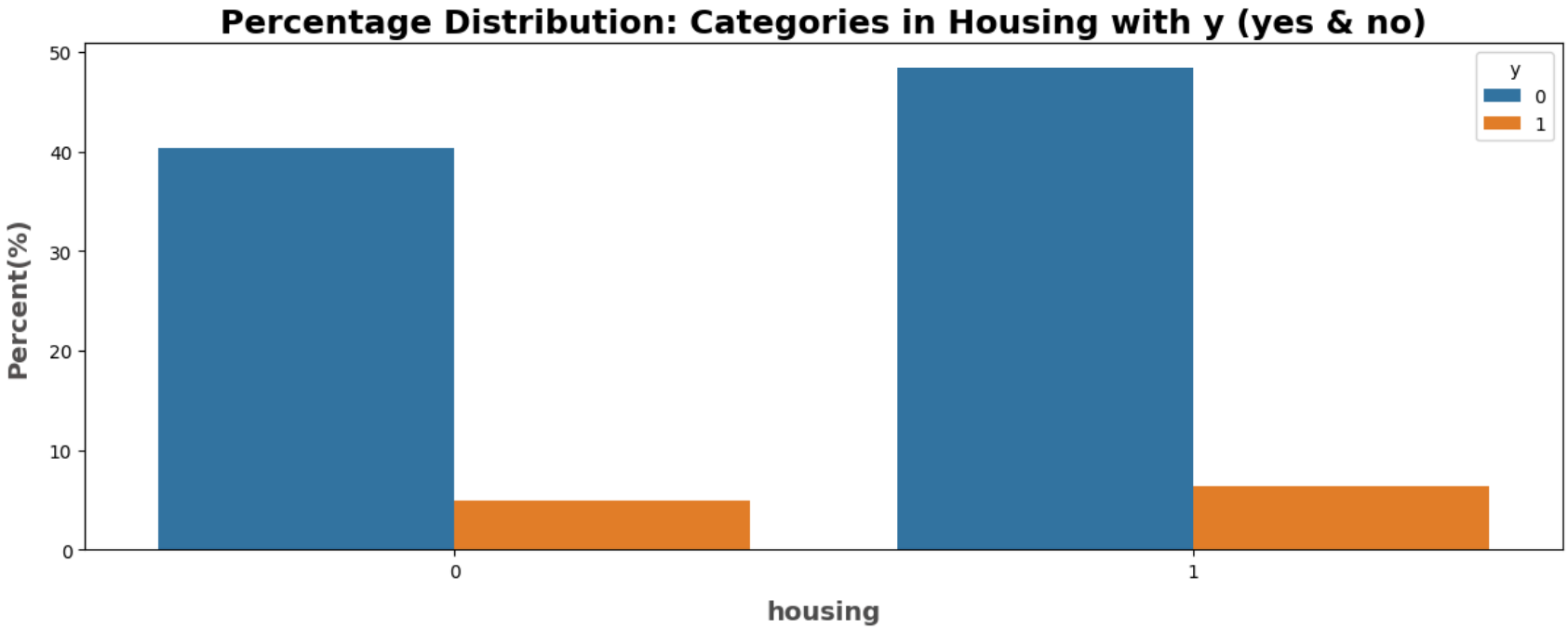
In [59]:
```
for i, col in enumerate(df.columns):
    if col in categoricalVariables:
        fig = plt.figure(figsize=(14.5,5))
        df[col+'_percent'] = df.groupby(['y', col])[col].transform(lambda x: len(x)*100) / len(df)
        sns.barplot(x=col, y=col+'_percent', hue='y', data=df).set(xlabel=col, ylabel='Percent(%)')
        plt.xlabel(xlabel=col,size=14, color='#4f4e4e',labelpad=10,fontweight='bold')
        plt.ylabel(ylabel='Percent(%)',size=14, color='#4f4e4e', labelpad=10, fontweight='bold')
        plotTitle = "Percentage Distribution: Categories in "+ col.capitalize() + " with y (yes & no)"
        plt.title(plotTitle,size=18, fontweight='bold')
        plt.show()
        df = df.drop(columns=[col+'_percent'])
```

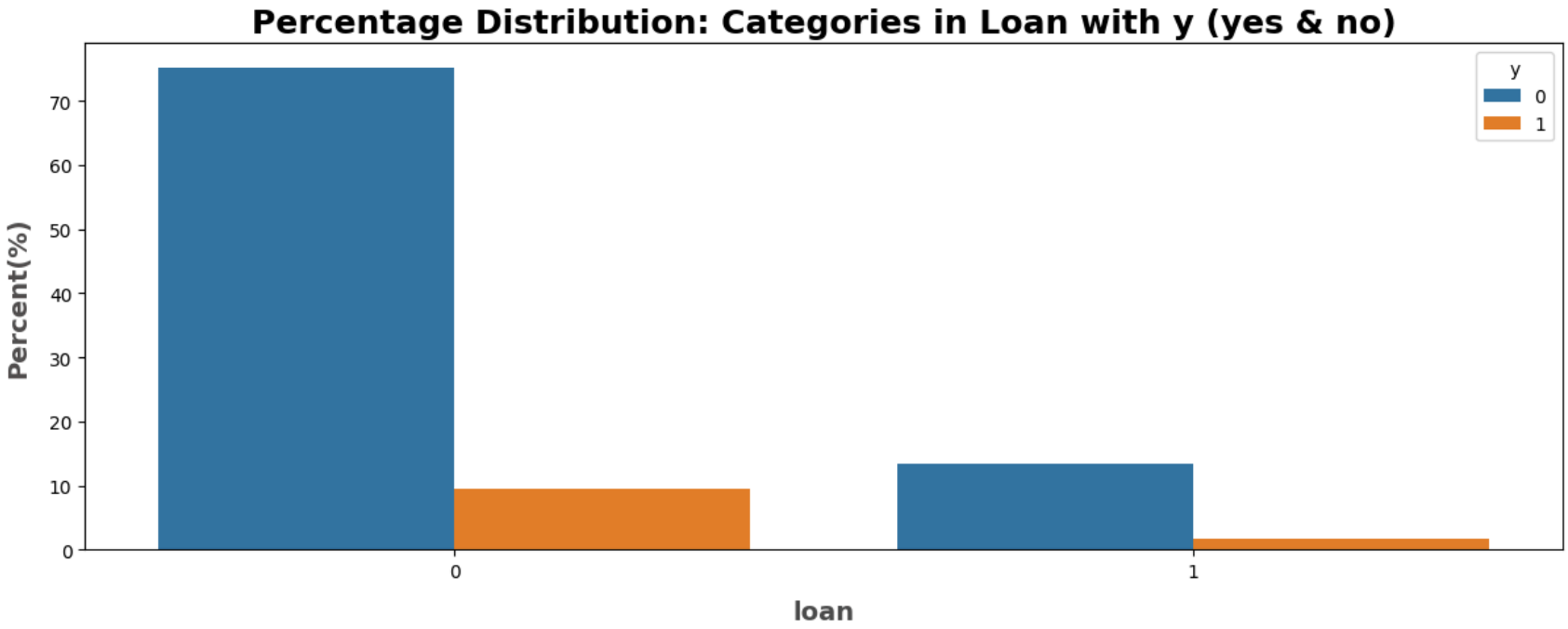


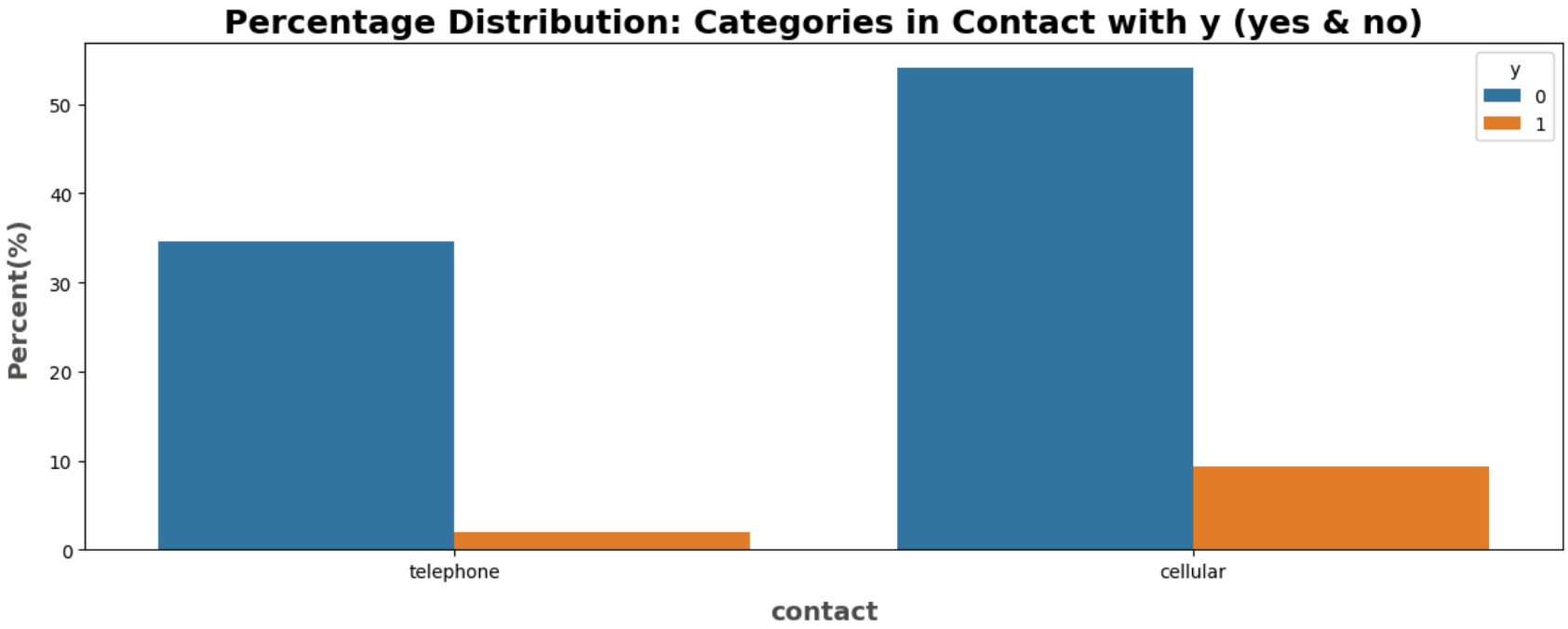

Percentage Distribution: Categories in Marital with y (yes & no)
Percent(%)
0
10
20
30
40
50
y
0
1
married
single
divorced
marital

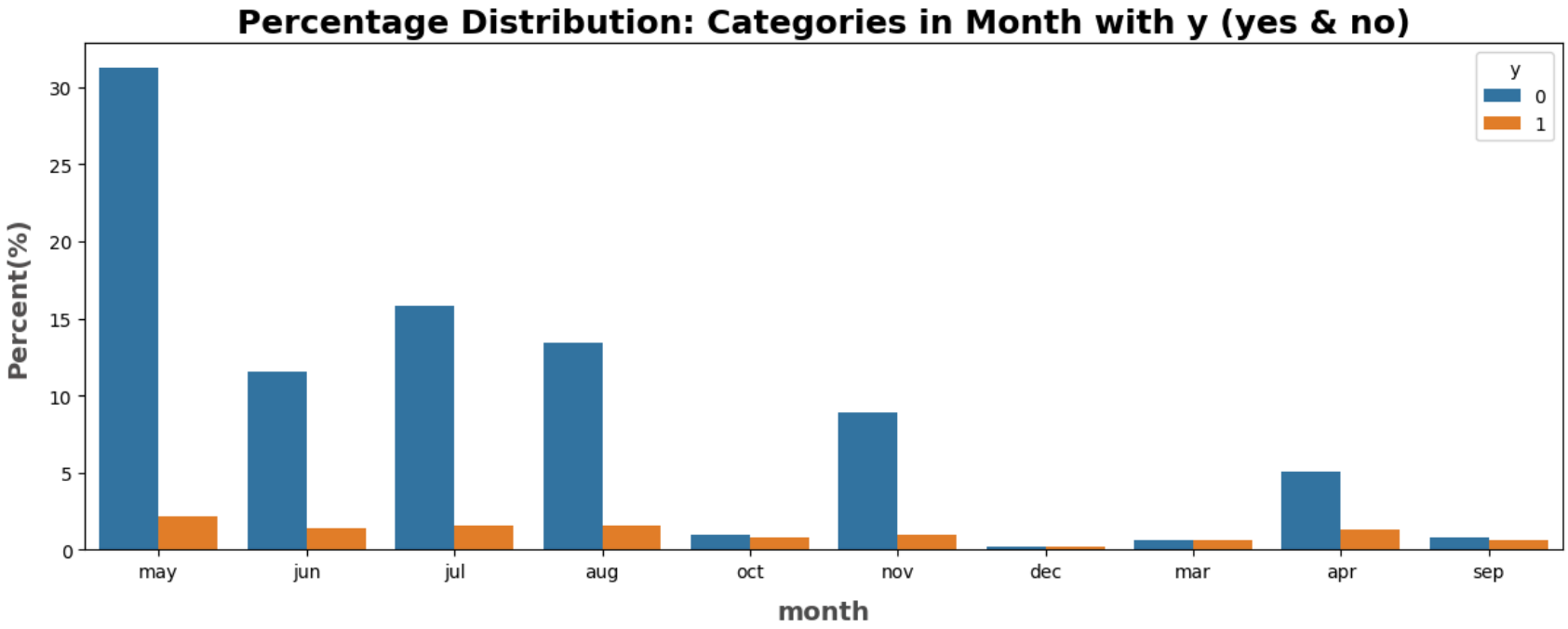

Percentage Distribution: Categories in Education with y (yes & no)
Percent(%)
0
5
10
15
20
25
30
y
0
1
basic.4y
high.school
basic.6y
basic.9y
professional.course
university.degree
illiterate
education

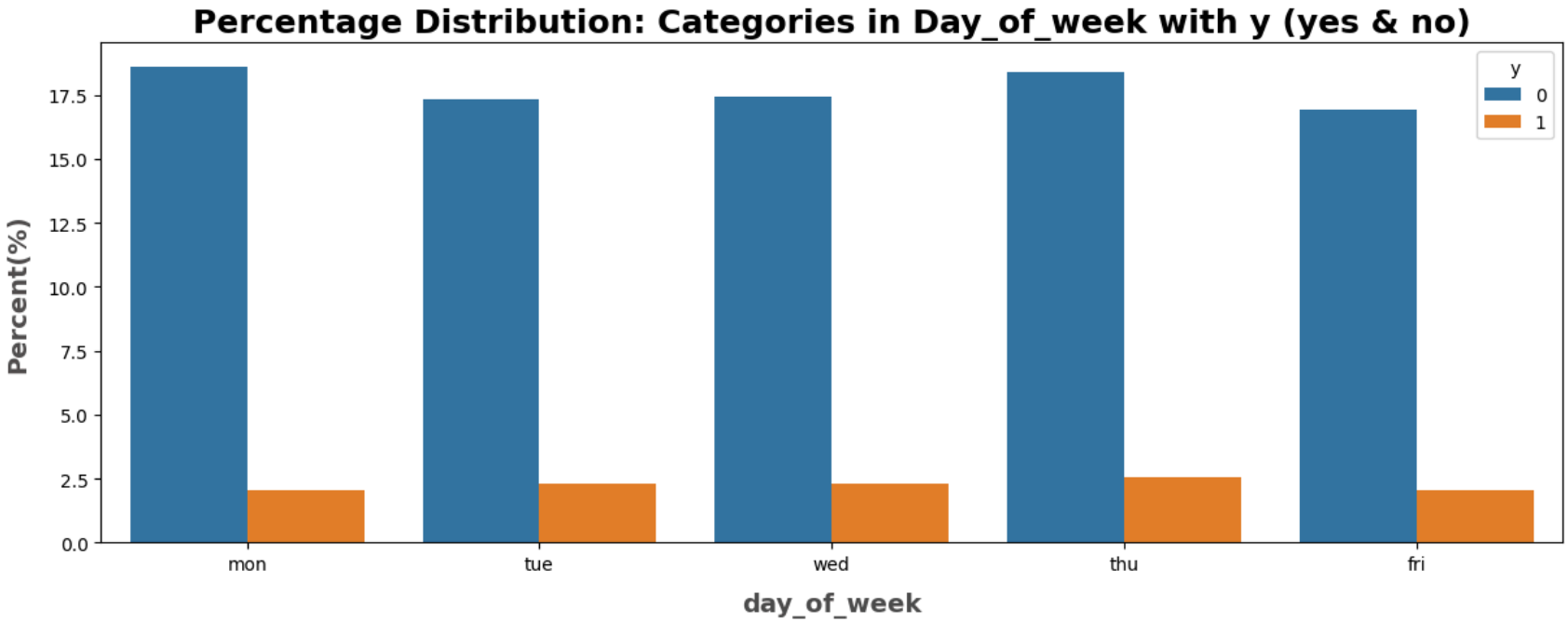

Percentage Distribution: Categories in Default with y (yes & no)
Percent(%)
0
20
40
60
80
y
0
1
0
1
default

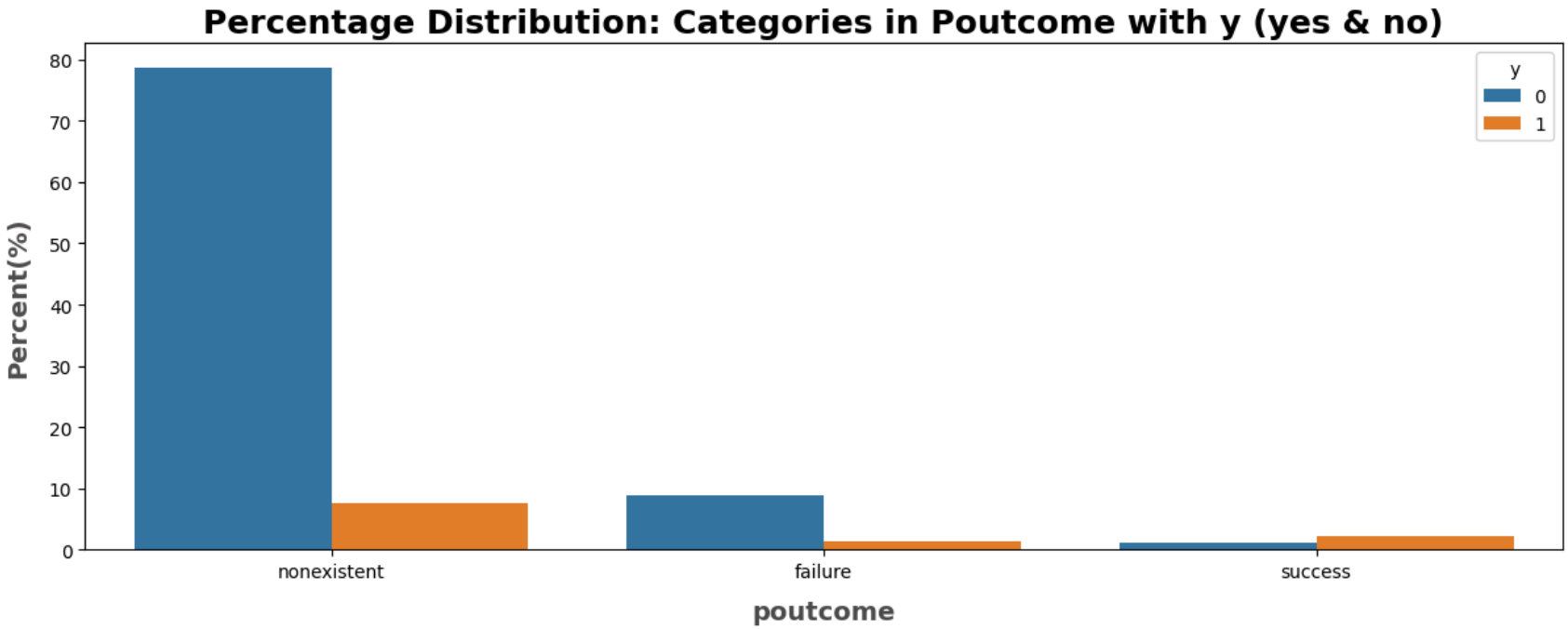

Percentage Distribution: Categories in Housing with y (yes & no)
Percent(%)
0
10
20
30
40
50
y
0
1
0
1
housing


Percentage Distribution: Categories in Loan with y (yes & no)
Percent(%)
0
10
20
30
40
50
60
70
0
1
loan
y
0
1


Percentage Distribution: Categories in Contact with y (yes & no)
Percent(%)
0
10
20
30
40
50
telephone
cellular
contact
y
0
1


Percentage Distribution: Categories in Month with y (yes & no)
Percent(%)
0
5
10
15
20
25
30
may
jun
jul
aug
oct
nov
dec
mar
apr
sep
month
y
0
1


Percentage Distribution: Categories in Day_of_week with y (yes & no)
Percent(%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
mon
tue
wed
thu
fri
day_of_week
y
0
1



Clean Data & Replace unknowns

In [23]:
```
attributesWithUnknown = ['job', 'marital', 'education', 'default', 'housing', 'loan']

for col in attributesWithUnknown:
    mostFrequent = df[col].mode()[0]
    df[col] = df[col].replace('unknown', mostFrequent)
```

In [24]:
```
# 999 means "not contacted" — replace with 0 or NaN before stats
df['pdays'] = df['pdays'].replace(999, 0)
```

In [25]:
```
# yes/no columns → 1/0
binary_cols = ['default', 'housing', 'loan', 'y']
for col in binary_cols:
    df[col] = df[col].map({'yes': 1, 'no': 0})
```

In [26]:
```
# Binning age
df['age_bin'] = pd.cut(df['age'], bins=[0,25,35,45,55,65,100],
```

```
                        labels=['<25','25-35','35-45','45-55','55-65','65+'])

# Binning pdays
df['pdays_bin'] = pd.cut(df['pdays'], bins=[-1,0,7,14,30,100],
                         labels=['not_contacted','1w','2w','1m','1m+'])
```

In [27]:
```
categorical_cols = ['contact', 'default', 'day_of_week', 'job', 'housing',
                    'education', 'loan', 'poutcome', 'month', 'marital']

df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
```

In [28]:
```
df_encoded
```

Out[28]:

	age	duration	campaign	pdays	previous	emp.var.rate	cons.price.idx	cons.conf.idx	euribor3m	nr.employed	.
0	56	261	1	0	0	1.1	93.994	-36.4	4.857	5191.0	.
1	57	149	1	0	0	1.1	93.994	-36.4	4.857	5191.0	.
2	37	226	1	0	0	1.1	93.994	-36.4	4.857	5191.0	.
3	40	151	1	0	0	1.1	93.994	-36.4	4.857	5191.0	.
4	56	307	1	0	0	1.1	93.994	-36.4	4.857	5191.0	.
...	...	...	...	...	...	...	...	...	...	...	.
41183	73	334	1	0	0	-1.1	94.767	-50.8	1.028	4963.6	.
41184	46	383	1	0	0	-1.1	94.767	-50.8	1.028	4963.6	.
41185	56	189	2	0	0	-1.1	94.767	-50.8	1.028	4963.6	.
41186	44	442	1	0	0	-1.1	94.767	-50.8	1.028	4963.6	.
41187	74	239	3	0	1	-1.1	94.767	-50.8	1.028	4963.6	.

41188 rows × 50 columns

Mean, Median, Standard Deviation & Correlation

In [29]:
```
df_encoded.select_dtypes(include='number').agg(['mean', 'median', 'std'])
```

Out[29]:

	age	duration	campaign	pdays	previous	emp.var.rate	cons.price.idx	cons.conf.idx	euribor3m	n
mean	40.02406	258.285010	2.567593	0.221229	0.172963	0.081886	93.575664	-40.502600	3.621291	5
median	38.00000	180.000000	2.000000	0.000000	0.000000	1.100000	93.749000	-41.800000	4.857000	5
std	10.42125	259.279249	2.770014	1.348874	0.494901	1.570960	0.578840	4.628198	1.734447	

3 rows × 48 columns

In [30]:
```
df_encoded.select_dtypes(include='number').corr()
```

Out[30]:

	age	duration	campaign	pdays	previous	emp.var.rate	cons.price.idx
age	1.000000	-0.000866	4.593580e-03	0.018132	0.024365	-0.000371	0.000857
duration	-0.000866	1.000000	-7.169923e-02	0.047853	0.020640	-0.027968	0.005312
campaign	0.004594	-0.071699	1.000000e+00	-0.041984	-0.079141	0.150754	0.127836
pdays	0.018132	0.047853	-4.198361e-02	1.000000	0.485182	-0.218942	-0.041787
previous	0.024365	0.020640	-7.914147e-02	0.485182	1.000000	-0.420489	-0.203130
emp.var.rate	-0.000371	-0.027968	1.507538e-01	-0.218942	-0.420489	1.000000	0.775334
cons.price.idx	0.000857	0.005312	1.278359e-01	-0.041787	-0.203130	0.775334	1.000000
cons.conf.idx	0.129372	-0.008173	-1.373310e-02	0.064192	-0.050936	0.196041	0.058986
euribor3m	0.010767	-0.032897	1.351325e-01	-0.251822	-0.454494	0.972245	0.688230
nr.employed	-0.017725	-0.044703	1.440949e-01	-0.323122	-0.501333	0.906970	0.522034
y	0.030399	0.405274	-6.635741e-02	0.267088	0.230181	-0.298334	-0.136211
contact_telephone	0.007021	-0.026657	7.736754e-02	-0.098173	-0.212848	0.393584	0.591474
default_1	0.001891	-0.005101	-3.802952e-03	-0.001400	0.002766	0.004445	-0.003141
day_of_week_mon	0.019027	-0.023068	1.440505e-02	0.002243	-0.002012	-0.018139	0.001273
day_of_week_thu	-0.021356	0.014666	5.141381e-03	0.006961	0.001512	-0.007673	-0.017143
day_of_week_tue	0.019137	0.004140	-2.512492e-02	0.003591	0.000090	0.014582	0.001216
day_of_week_wed	-0.023671	0.014537	-2.108788e-02	-0.002644	-0.003929	0.026797	0.012479
job_blue-collar	-0.024191	0.012992	-1.774573e-03	-0.053833	-0.054845	0.057264	0.075322

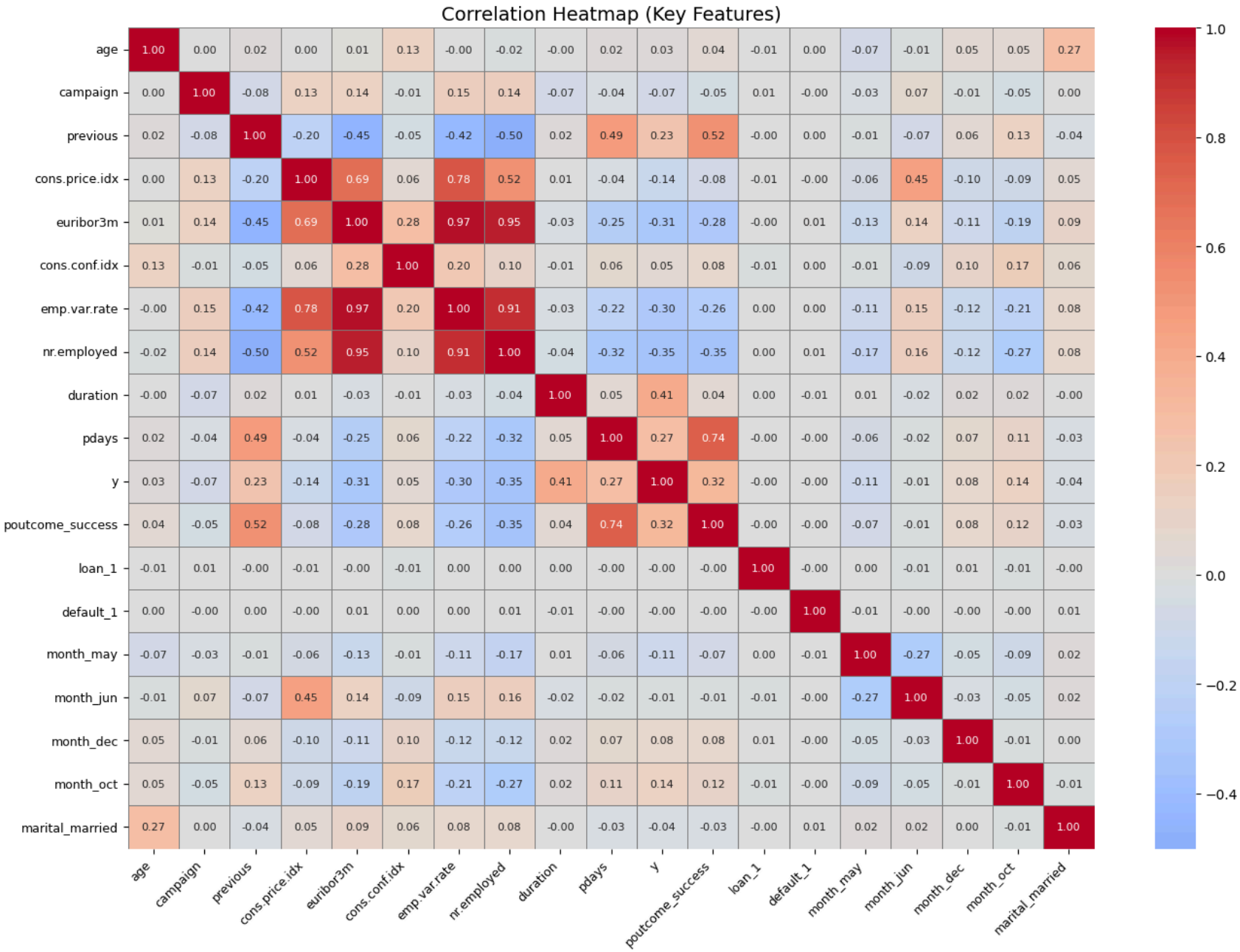
	age	duration	campaign	pdays	previous	emp.var.rate	cons.price.idx
job_entrepreneur	0.031213	0.003679	-2.203068e-03	-0.016289	-0.013239	0.009363	0.009825
job_housemaid	0.085403	-0.004908	4.226363e-03	-0.000171	-0.011569	0.036367	0.028335
job_management	0.062040	-0.001308	-9.134637e-03	-0.003496	0.006735	-0.016642	-0.025268
job_retired	0.440772	0.012421	-6.846710e-03	0.055292	0.065113	-0.103678	-0.052251
job_self-employed	-0.001356	0.004270	6.360926e-03	-0.014435	-0.011231	0.001477	-0.005122
job_services	-0.065731	0.000142	2.382935e-03	-0.028550	-0.011885	0.019431	0.033282
job_student	-0.199749	0.014432	-2.465706e-02	0.091301	0.104671	-0.139725	-0.062117
job_technician	-0.064383	-0.013742	1.556883e-03	0.000985	-0.017142	0.054268	-0.010849
job_unemployed	-0.004426	-0.005413	-2.001648e-04	0.015759	0.008426	-0.019581	-0.003262
housing_1	-0.001755	-0.008334	-1.095916e-02	0.003969	0.021677	-0.059100	-0.077170
education_basic.6y	0.009897	0.005739	-9.911340e-04	-0.019948	-0.021493	0.023065	0.031201
education_basic.9y	-0.038320	0.004810	-5.278159e-03	-0.028145	-0.026418	0.020422	0.034202
education_high.school	-0.106550	0.005500	1.945928e-04	-0.002221	0.014347	-0.017078	0.008704
education_illiterate	0.017007	0.001491	-2.187709e-03	0.001738	-0.002613	-0.002865	-0.009332
education_professional.course	0.002054	-0.008472	2.553815e-03	0.009564	-0.007631	0.022154	-0.003827
education_university.degree	-0.039154	-0.010790	5.895741e-07	0.029850	0.034133	-0.045028	-0.075987
loan_1	-0.007198	0.000121	5.294371e-03	-0.002772	-0.002143	0.001193	-0.005944
poutcome_nonexistent	-0.016376	-0.011665	8.764482e-02	-0.412396	-0.878776	0.472501	0.304264

	age	duration	campaign	pdays	previous	emp.var.rate	cons.price.idx
poutcome_success	0.035626	0.044876	-5.089315e-02	0.738822	0.524045	-0.256886	-0.077416
month_aug	0.065309	-0.041889	1.500854e-02	0.000869	-0.048852	0.177895	-0.192027
month_dec	0.046563	0.022438	-9.818916e-03	0.065873	0.059557	-0.124174	-0.099017
month_jul	-0.043563	0.026819	9.838453e-02	-0.030189	-0.116785	0.314863	0.253303
month_jun	-0.009532	-0.023351	7.014425e-02	-0.020102	-0.069905	0.148653	0.445513
month_mar	0.008390	-0.005516	-1.524356e-02	0.077621	0.072733	-0.138849	-0.095783
month_may	-0.067523	0.008054	-3.270422e-02	-0.056398	-0.012534	-0.111356	-0.057274
month_nov	0.030045	-0.015941	-7.902830e-02	0.003170	0.081050	-0.105917	-0.215273
month_oct	0.053008	0.015454	-4.912621e-02	0.111835	0.127754	-0.213706	-0.092174
month_sep	0.040889	0.018011	-3.395251e-02	0.127714	0.157651	-0.170366	-0.046905
marital_married	0.271596	-0.003210	3.224140e-03	-0.029798	-0.042638	0.079490	0.045747
marital_single	-0.411703	0.007808	-7.623924e-03	0.039683	0.048485	-0.099403	-0.063013

48 rows × 48 columns

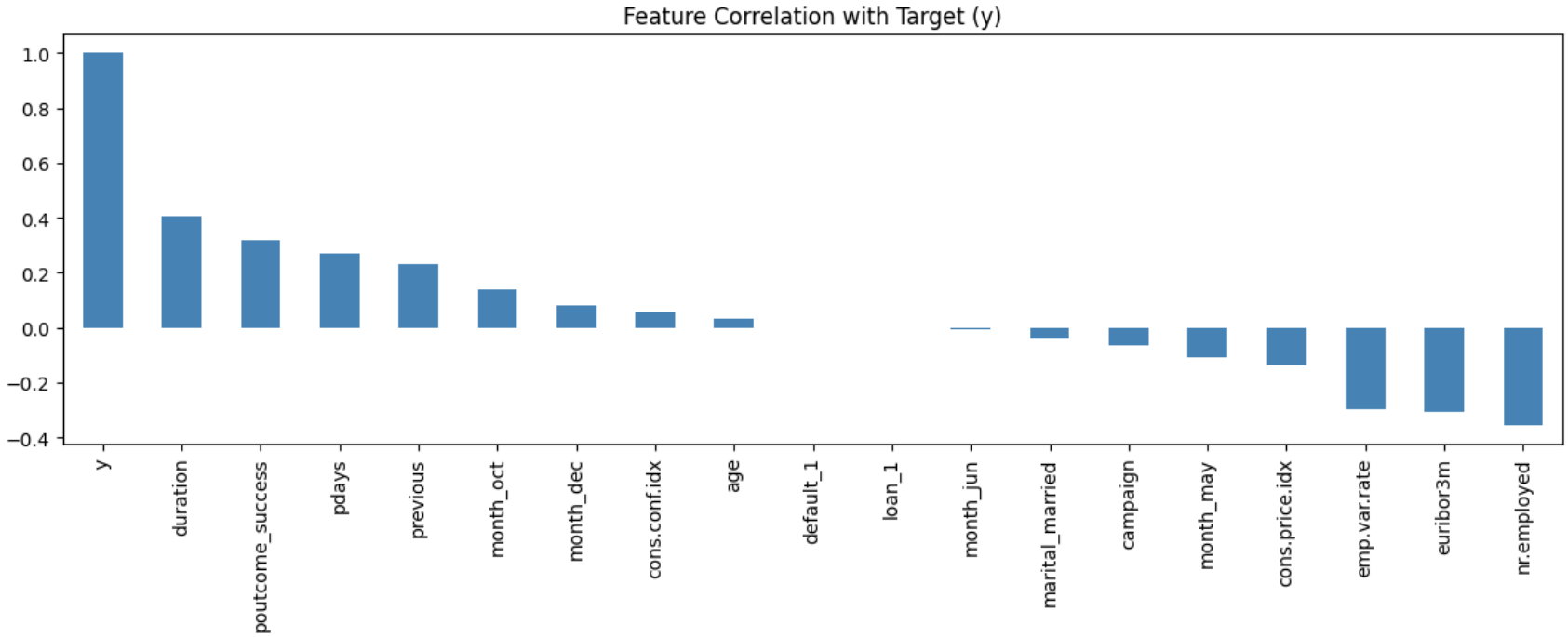
In [32]:
```
# Keep only original numerical columns + target + key encoded features
cols_to_keep = [
    # Core numerical
    'age', 'campaign', 'previous', 'cons.price.idx', 'euribor3m',
    'cons.conf.idx', 'emp.var.rate', 'nr.employed', 'duration', 'pdays',
    # Target
    'y',
    # Key encoded (only the most meaningful ones)
```

```
    'poutcome_success', 'loan_1', 'default_1',
    'month_may', 'month_jun', 'month_dec', 'month_oct',
    'marital_married'
]

# Filter only columns that exist in your df
cols_to_keep = [c for c in cols_to_keep if c in df_encoded.columns]
df_corr = df_encoded[cols_to_keep].corr()

# Plot
plt.figure(figsize=(14, 10))
sns.heatmap(
    df_corr,
    annot=True,
    fmt='.2f',          # 2 decimal places
    cmap='coolwarm',
    center=0,
    linewidths=0.5,
    linecolor='gray',
    annot_kws={'size': 8}
)
plt.title('Correlation Heatmap (Key Features)', fontsize=14)
plt.xticks(rotation=45, ha='right', fontsize=9)
plt.yticks(fontsize=9)
plt.tight_layout()
plt.show()
```



In [33]:
```
## For simpler view
df_encoded[cols_to_keep].corr()['y'].sort_values(ascending=False).plot(
```

```
    kind='bar', figsize=(12, 5), title='Feature Correlation with Target (y)', color='steelblue'
)
plt.tight_layout()
plt.show()
```

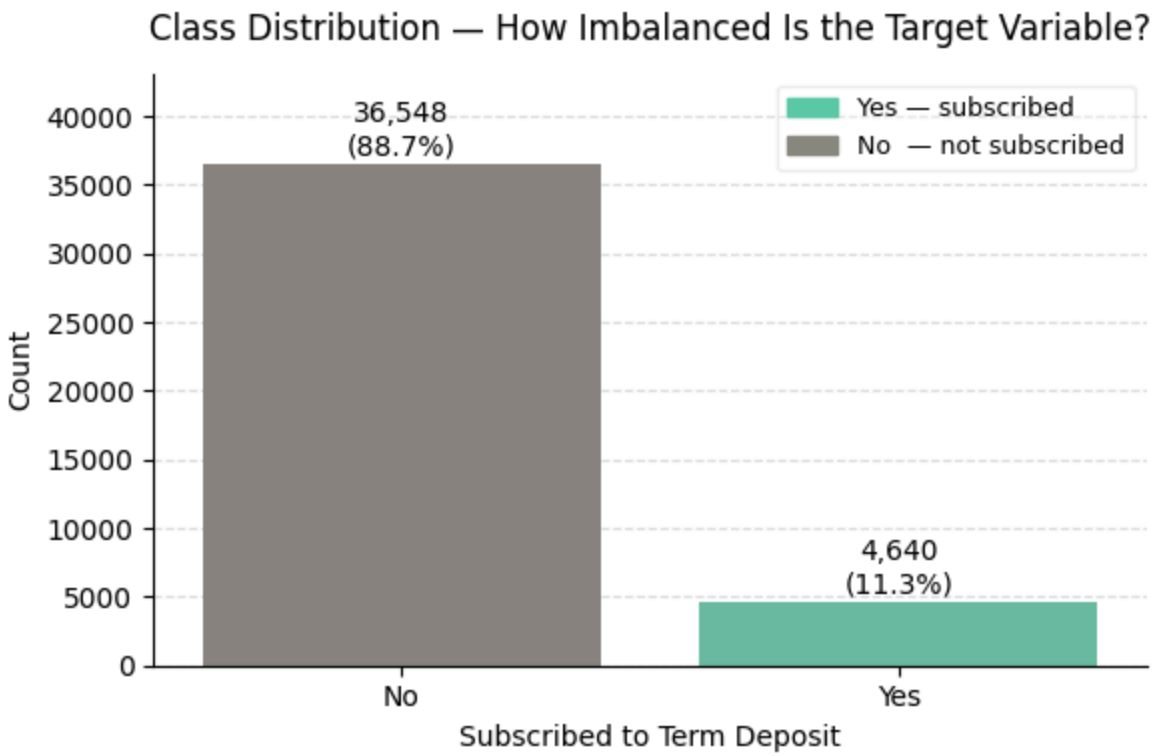


Data Visualization

Class Distribution

In [40]:
```
df_encoded['y_label'] = df_encoded['y'].map({0: 'No', 1: 'Yes'})

counts = df_encoded['y_label'].value_counts()
total = len(df_encoded)

fig, ax = plt.subplots(figsize=(6, 4))

sns.countplot(
    x='y_label',
```

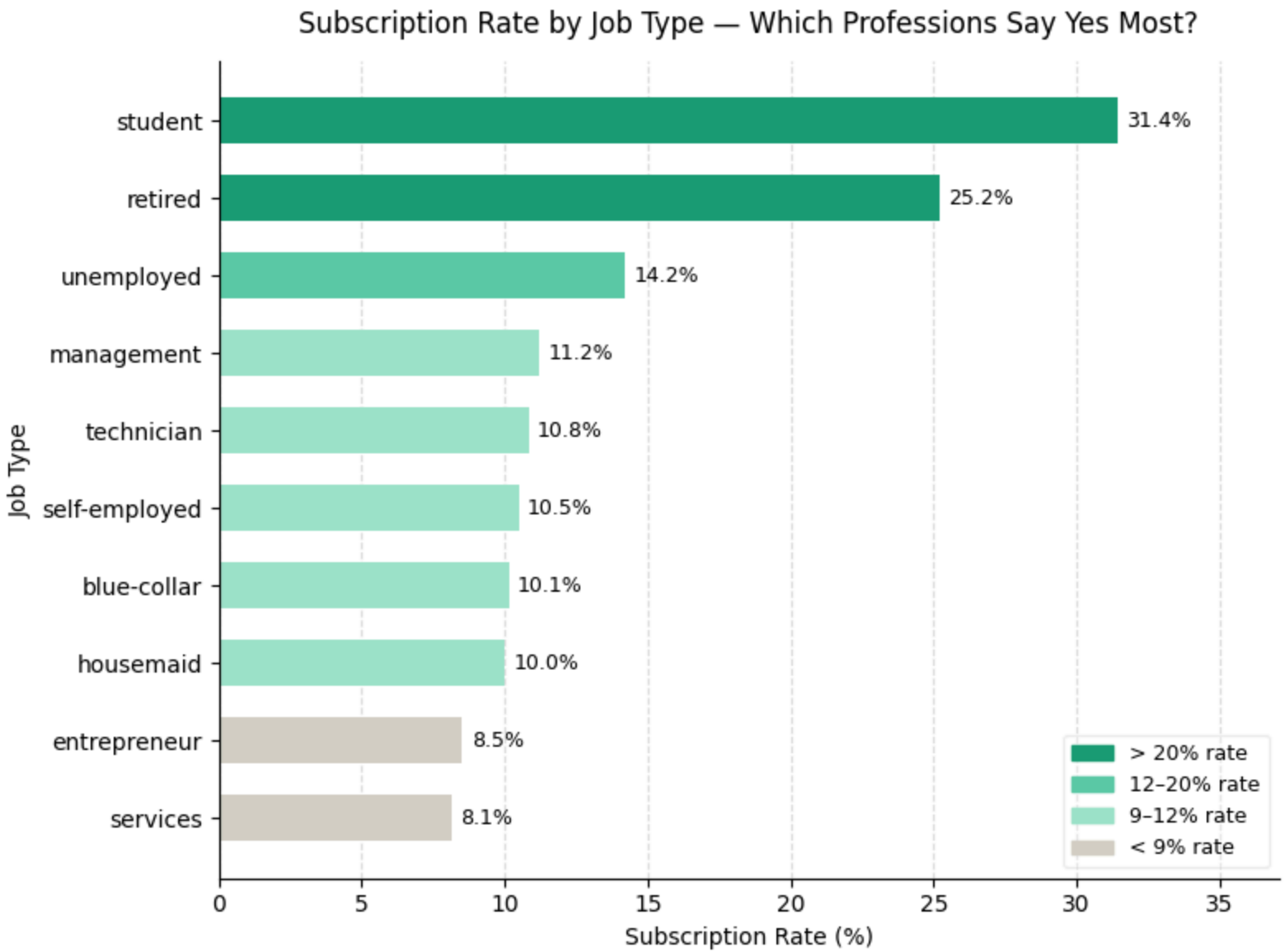
```
    data=df_encoded,
    order=['No', 'Yes'],
    palette={'No': '#888780', 'Yes': '#5DCAA5'},
    ax=ax
)

# Add count + percentage labels on bars
for bar, label in zip(ax.patches, ['No', 'Yes']):
    count = counts.get(label, 0)
    pct = count / total * 100
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 200,
        f'{count:,}\n({pct:.1f}%)',
        ha='center', va='bottom', fontsize=10
    )

# Styling
ax.set_title('Class Distribution — How Imbalanced Is the Target Variable?', fontsize=12, pad=12)
ax.set_xlabel('Subscribed to Term Deposit')
ax.set_ylabel('Count')
ax.set_xticklabels(['No', 'Yes'])
ax.spines[['top', 'right']].set_visible(False)
ax.yaxis.grid(True, linestyle='--', alpha=0.4)
ax.set_axisbelow(True)
ax.set_ylim(0, counts.max() * 1.18)

legend = [
    mpatches.Patch(color='#5DCAA5', label=f"Yes — subscribed"),
    mpatches.Patch(color='#888780', label=f"No  — not subscribed")
]
ax.legend(handles=legend, fontsize=9, framealpha=0.3)

plt.tight_layout()
plt.show()
```



Subscription rate by job type — which professions say yes most?

In [43]:
```
# Calculate subscription rate by job
job_cols = [c for c in df_encoded.columns if c.startswith('job_')]
df_encoded['job'] = df_encoded[job_cols].idxmax(axis=1).str.replace('job_', '')
job_sub = df_encoded.groupby('job')['y'].mean() * 100
job_sub_rate = job_sub.sort_values(ascending=True)

# Colors based on rate
colors = ['#1D9E75' if r > 20 else '#5DCAA5' if r > 12 else '#9FE1CB' if r > 9 else '#D3D1C7'
          for r in job_sub_rate]

fig, ax = plt.subplots(figsize=(8, 6))

bars = ax.barh(
    job_sub_rate.index,
    job_sub_rate.values,
```

```
    color=colors,
    edgecolor='none',
    height=0.6
)

# Add percentage labels on bars
for bar, val in zip(bars, job_sub_rate.values):
    ax.text(
        bar.get_width() + 0.3,
        bar.get_y() + bar.get_height() / 2,
        f'{val:.1f}%',
        va='center', fontsize=9
    )

# Styling
ax.set_title('Subscription Rate by Job Type — Which Professions Say Yes Most?',
             fontsize=12, pad=12)
ax.set_xlabel('Subscription Rate (%)')
ax.set_ylabel('Job Type')
ax.set_xlim(0, job_sub_rate.max() * 1.18)
ax.spines[['top', 'right']].set_visible(False)
ax.xaxis.grid(True, linestyle='--', alpha=0.4)
ax.set_axisbelow(True)

# Legend
import matplotlib.patches as mpatches
legend = [
    mpatches.Patch(color='#1D9E75', label='> 20% rate'),
    mpatches.Patch(color='#5DCAA5', label='12–20% rate'),
    mpatches.Patch(color='#9FE1CB', label='9–12% rate'),
    mpatches.Patch(color='#D3D1C7', label='< 9% rate'),
]
ax.legend(handles=legend, fontsize=9, framealpha=0.3, loc='lower right')

plt.tight_layout()
plt.show()
```

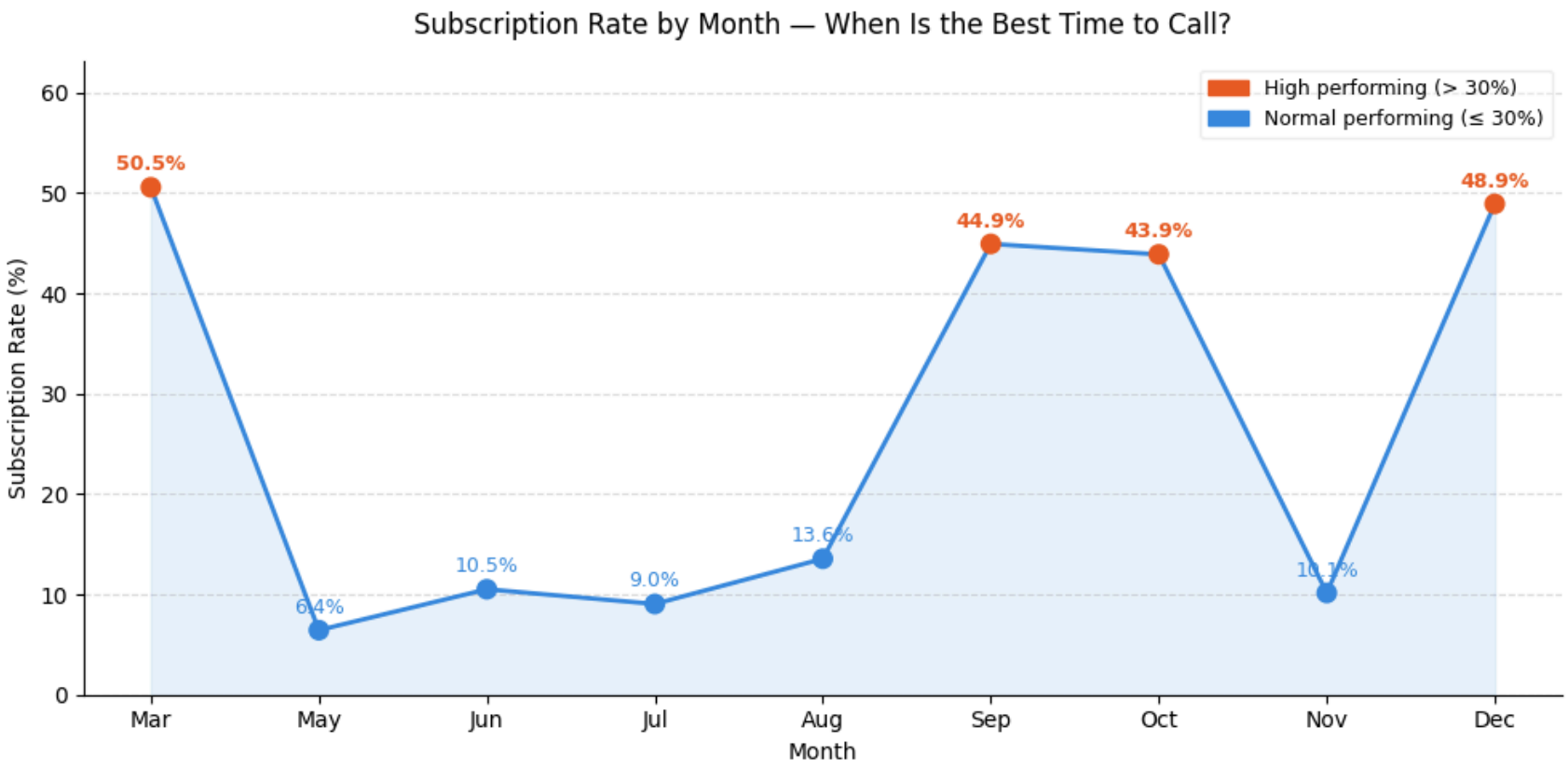


Subscription rate by month — when is the best time to call?

In [45]:
```
# Recover month column from one-hot encoded columns
month_cols = [c for c in df_encoded.columns if c.startswith('month_')]
df_encoded['month'] = df_encoded[month_cols].idxmax(axis=1).str.replace('month_', '')
```

```
# Month order
month_order = ['jan','feb','mar','apr','may','jun',
                'jul','aug','sep','oct','nov','dec']

# Subscription rate by month
month_sub = df_encoded.groupby('month')['y'].mean() * 100
month_sub_rate = month_sub.reindex(month_order).dropna()

# Color: highlight high-performing months
colors = ['#E85D24' if r > 30 else '#378ADD' for r in month_sub_rate.values]

fig, ax = plt.subplots(figsize=(10, 5))

# Line plot
ax.plot(
    month_sub_rate.index,
    month_sub_rate.values,
    color='#378ADD',
    linewidth=2,
    zorder=1
)

# Shaded area under line
ax.fill_between(
    month_sub_rate.index,
    month_sub_rate.values,
    alpha=0.1,
    color='#378ADD'
)

# Scatter points (colored by performance)
ax.scatter(
    month_sub_rate.index,
    month_sub_rate.values,
    color=colors,
    s=70,
    zorder=2
)

# Add percentage labels above each point
for month, val, color in zip(month_sub_rate.index, month_sub_rate.values, colors):
```

```
    ax.text(
        month, val + 1.2,
        f'{val:.1f}%',
        ha='center', va='bottom',
        fontsize=9,
        color=color,
        fontweight='bold' if color == '#E85D24' else 'normal'
    )

# Styling
ax.set_title('Subscription Rate by Month — When Is the Best Time to Call?',
             fontsize=12, pad=12)
ax.set_xlabel('Month')
ax.set_ylabel('Subscription Rate (%)')
ax.set_xticks(range(len(month_sub_rate.index)))
ax.set_xticklabels([m.capitalize() for m in month_sub_rate.index])
ax.spines[['top', 'right']].set_visible(False)
ax.yaxis.grid(True, linestyle='--', alpha=0.4)
ax.set_axisbelow(True)
ax.set_ylim(0, month_sub_rate.max() * 1.25)

# Legend
legend = [
    mpatches.Patch(color='#E85D24', label='High performing (> 30%)'),
    mpatches.Patch(color='#378ADD', label='Normal performing (≤ 30%)'),
]
ax.legend(handles=legend, fontsize=9, framealpha=0.3)

plt.tight_layout()
plt.show()

# Drop temp column to keep df_encoded clean
df_encoded.drop(columns='month', inplace=True)
```

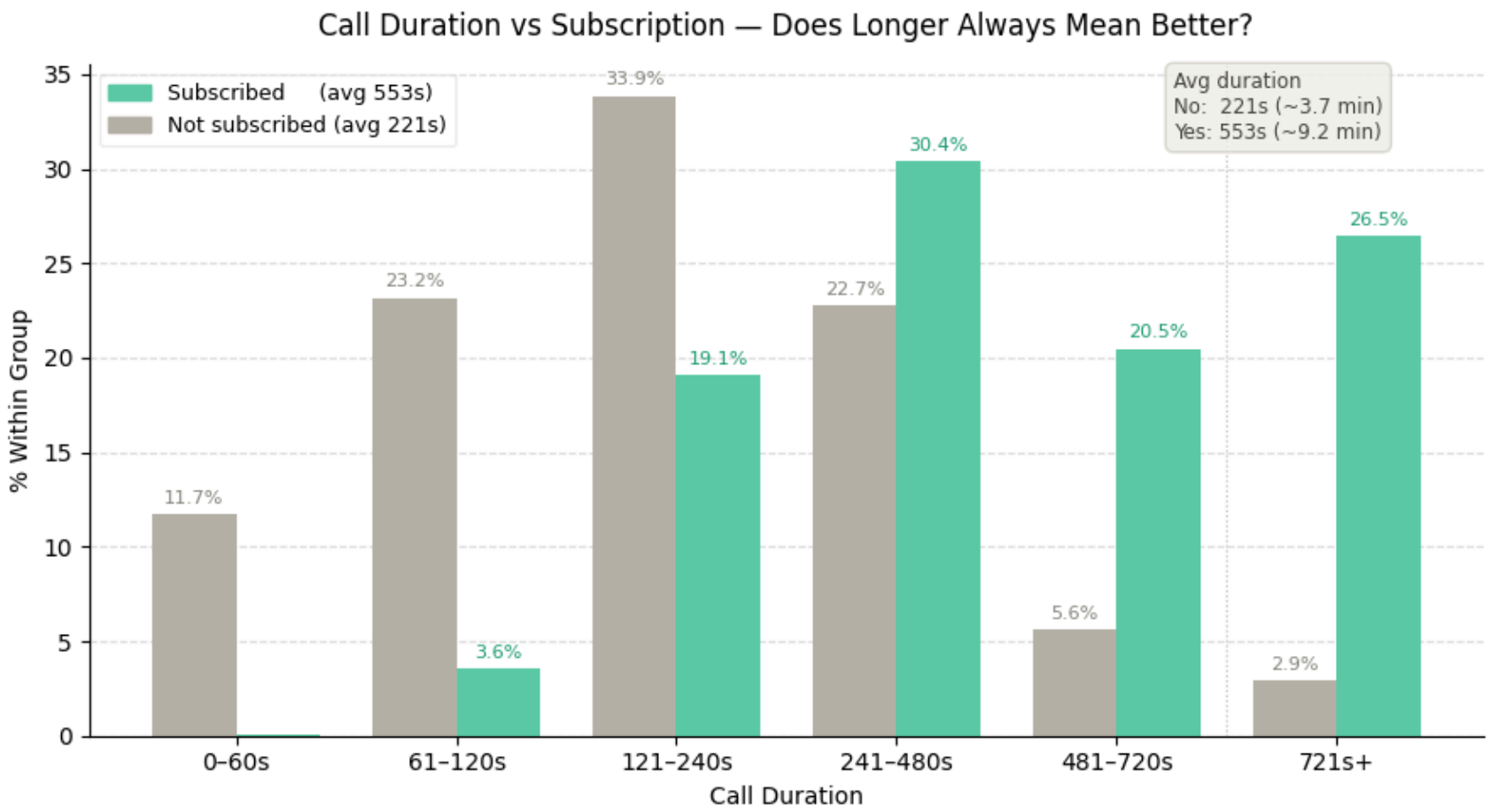


Call duration vs subscription — does longer always mean better?

In [46]:
```
# Define duration bins
bins   = [0, 60, 120, 240, 480, 720, float('inf')]
labels = ['0–60s', '61–120s', '121–240s', '241–480s', '481–720s', '721s+']

# Bin the duration column
df_encoded['duration_bin'] = pd.cut(
    df_encoded['duration'],
    bins=bins,
    labels=labels,
    right=True
)

# Calculate % distribution within each class
```

```
no_dist  = (df_encoded[df_encoded['y'] == 0]['duration_bin']
            .value_counts(normalize=True)
            .reindex(labels) * 100)

yes_dist = (df_encoded[df_encoded['y'] == 1]['duration_bin']
            .value_counts(normalize=True)
            .reindex(labels) * 100)

# Average duration per class
avg_no  = df_encoded[df_encoded['y'] == 0]['duration'].mean()
avg_yes = df_encoded[df_encoded['y'] == 1]['duration'].mean()

# --- Plot ---
fig, ax = plt.subplots(figsize=(9, 5))

x     = np.arange(len(labels))
width = 0.38

bars_no  = ax.bar(x - width/2, no_dist.values,  width, color='#B4B2A9',
                  label='Not subscribed', edgecolor='none', zorder=2)
bars_yes = ax.bar(x + width/2, yes_dist.values, width, color='#5DCAA5',
                  label='Subscribed',     edgecolor='none', zorder=2)

# Value labels on bars
for bar in bars_no:
    h = bar.get_height()
    if h > 1:
        ax.text(bar.get_x() + bar.get_width()/2, h + 0.4,
                f'{h:.1f}%', ha='center', va='bottom', fontsize=8, color='#888780')

for bar in bars_yes:
    h = bar.get_height()
    if h > 1:
        ax.text(bar.get_x() + bar.get_width()/2, h + 0.4,
                f'{h:.1f}%', ha='center', va='bottom', fontsize=8, color='#1D9E75')

# Styling
ax.set_title('Call Duration vs Subscription — Does Longer Always Mean Better?',
             fontsize=12, pad=12)
ax.set_xlabel('Call Duration')
ax.set_ylabel('% Within Group')
ax.set_xticks(x)
```

```
ax.set_xticklabels(labels)
ax.spines[['top', 'right']].set_visible(False)
ax.yaxis.grid(True, linestyle='--', alpha=0.4)
ax.set_axisbelow(True)

# Avg duration annotations
ax.axvline(x=4.5, color='gray', linestyle=':', linewidth=0.8, alpha=0.5)
fig.text(0.78, 0.82, f'Avg duration\nNo:  {avg_no:.0f}s (~{avg_no/60:.1f} min)\nYes: {avg_yes:.0f}s (~{avg_
         fontsize=8.5, color='#444441',
         bbox=dict(boxstyle='round,pad=0.4', facecolor='#F1EFE8', edgecolor='#D3D1C7', alpha=0.8))

# Legend
legend = [
    mpatches.Patch(color='#5DCAA5', label=f'Subscribed     (avg {avg_yes:.0f}s)'),
    mpatches.Patch(color='#B4B2A9', label=f'Not subscribed (avg {avg_no:.0f}s)'),
]
ax.legend(handles=legend, fontsize=9, framealpha=0.3)

plt.tight_layout()
plt.show()

# Drop temp column
df_encoded.drop(columns='duration_bin', inplace=True)
```

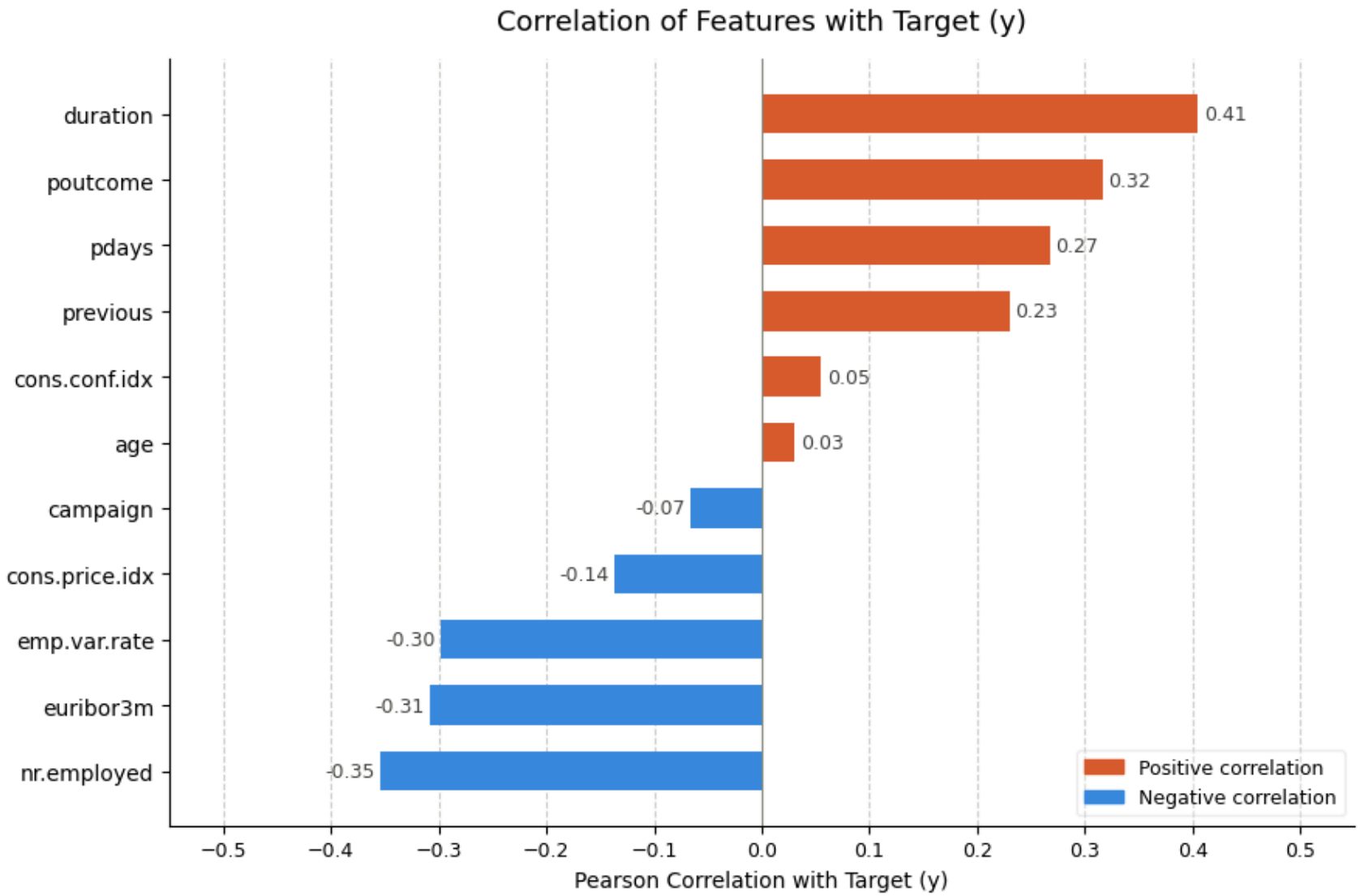


Correlation heatmap — key features vs target

In [48]:
```
# Select only core numerical features
cols_numerical = [
    'duration', 'previous', 'cons.conf.idx', 'age',
    'campaign', 'cons.price.idx', 'nr.employed',
    'euribor3m', 'emp.var.rate', 'pdays'
]

# Add poutcome_success if exists
extra_cols = ['poutcome_success']
cols_numerical += [c for c in extra_cols if c in df_encoded.columns]
```

```python
# Compute correlation with target y
corr_with_y = (df_encoded[cols_numerical + ['y']]
               .corr()['y']
               .drop('y')
               .sort_values(ascending=True))

# Colors: positive = coral/brown, negative = blue
colors = ['#D85A30' if v >= 0 else '#378ADD' for v in corr_with_y.values]

# Rename poutcome_success to poutcome for clean label
corr_with_y.index = [i.replace('poutcome_success', 'poutcome') for i in corr_with_y.index]

# --- Plot ---
fig, ax = plt.subplots(figsize=(9, 6))

bars = ax.barh(
    corr_with_y.index,
    corr_with_y.values,
    color=colors,
    edgecolor='none',
    height=0.6
)

# Add value labels on bars
for bar, val in zip(bars, corr_with_y.values):
    x_pos = bar.get_width() + 0.005 if val >= 0 else bar.get_width() - 0.005
    ha = 'left' if val >= 0 else 'right'
    ax.text(x_pos,
            bar.get_y() + bar.get_height() / 2,
            f'{val:.2f}',
            va='center', ha=ha, fontsize=9,
            color='#444441')

# Reference line at 0
ax.axvline(0, color='#888780', linewidth=0.8, linestyle='-')

# Styling
ax.set_title('Correlation of Features with Target (y)',
             fontsize=13, pad=14)
ax.set_xlabel('Pearson Correlation with Target (y)', fontsize=10)
ax.set_xlim(-0.55, 0.55)
```

```
ax.set_xticks([-0.5, -0.4, -0.3, -0.2, -0.1, 0, 0.1, 0.2, 0.3, 0.4, 0.5])
ax.spines[['top', 'right']].set_visible(False)
ax.xaxis.grid(True, linestyle='--', alpha=0.4, color='#888780')
ax.set_axisbelow(True)
ax.tick_params(axis='y', labelsize=10)
ax.tick_params(axis='x', labelsize=9)

# Legend
import matplotlib.patches as mpatches
legend = [
    mpatches.Patch(color='#D85A30', label='Positive correlation'),
    mpatches.Patch(color='#378ADD', label='Negative correlation'),
]
ax.legend(handles=legend, fontsize=9, framealpha=0.3, loc='lower right')

plt.tight_layout()
plt.show()
```



Previous outcome vs subscription — does past behavior predict future?

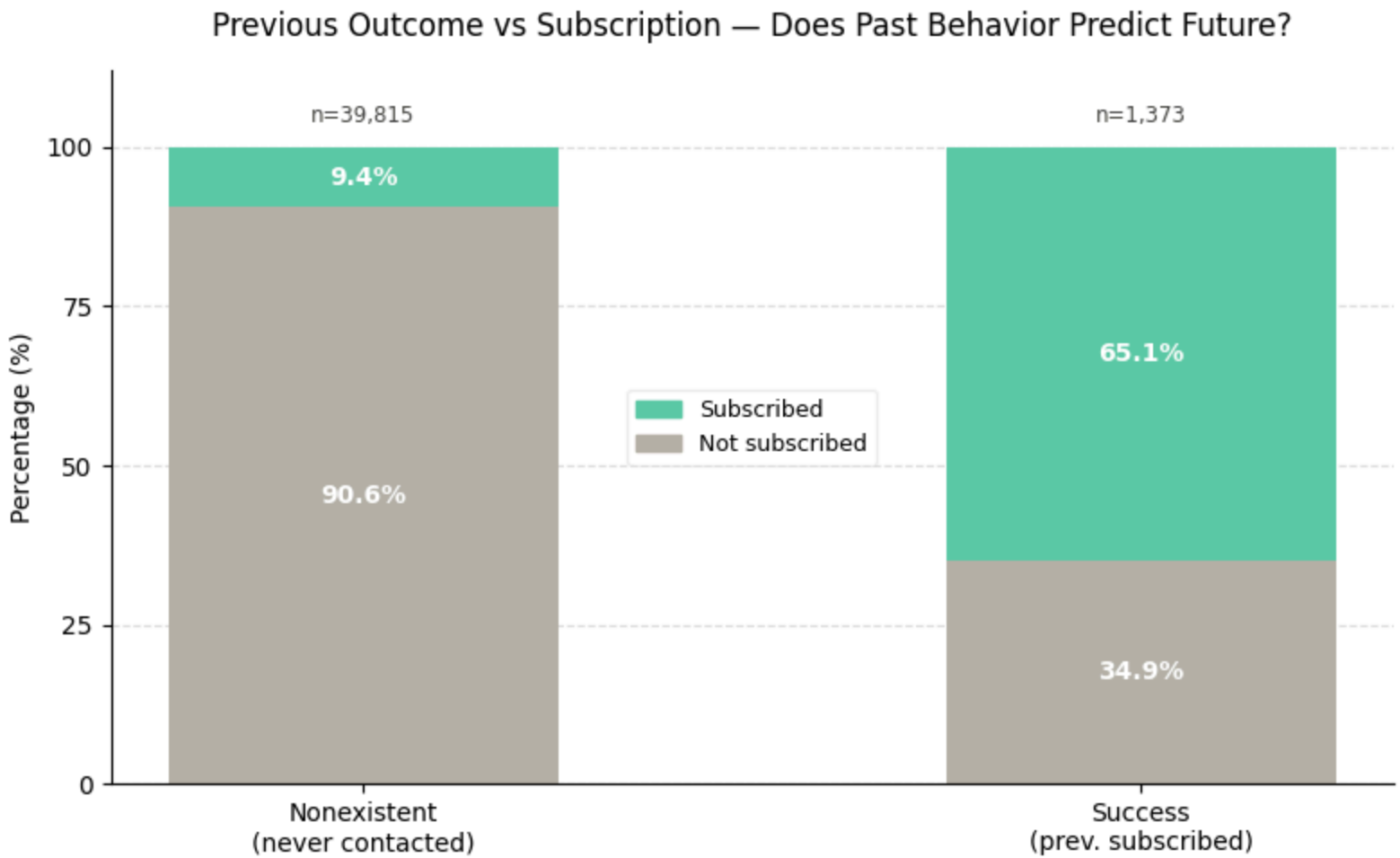
In [51]:
```
# Recover poutcome from one-hot encoded columns
poutcome_cols = [c for c in df_encoded.columns if c.startswith('poutcome_')]
print("Poutcome columns found:", poutcome_cols)

df_encoded['poutcome'] = df_encoded[poutcome_cols].idxmax(axis=1).str.replace('poutcome_', '')
```

```
# Calculate subscription rate per poutcome group
poutcome_group = df_encoded.groupby('poutcome')['y'].agg(
    total='count',
    subscribed='sum'
)
poutcome_group['not_subscribed'] = poutcome_group['total'] - poutcome_group['subscribed']
poutcome_group['sub_rate']       = (poutcome_group['subscribed'] / poutcome_group['total'] * 100).round(1)
poutcome_group['no_rate']        = (poutcome_group['not_subscribed'] / poutcome_group['total'] * 100).roun

print("\nSubscription rate by previous outcome:")
print(poutcome_group[['total', 'subscribed', 'not_subscribed', 'sub_rate']])

# Define display order and clean labels
order       = ['nonexistent', 'failure', 'success']
order       = [o for o in order if o in poutcome_group.index]
clean_labels = {
    'nonexistent': 'Nonexistent\n(never contacted)',
    'failure'    : 'Failure\n(prev. declined)',
    'success'    : 'Success\n(prev. subscribed)'
}

sub_rates = [poutcome_group.loc[o, 'sub_rate'] for o in order]
no_rates  = [poutcome_group.loc[o, 'no_rate']  for o in order]
x_labels  = [clean_labels.get(o, o) for o in order]

# --- Plot ---
fig, ax = plt.subplots(figsize=(8, 5))

x     = np.arange(len(order))
width = 0.5

bars_no  = ax.bar(x, no_rates,  width, label='Not subscribed',
                  color='#B4B2A9', edgecolor='none', zorder=2)
bars_yes = ax.bar(x, sub_rates, width, label='Subscribed',
                  bottom=no_rates, color='#5DCAA5',
                  edgecolor='none', zorder=2)

# Labels inside bars — Not subscribed
for bar, val in zip(bars_no, no_rates):
    if val > 5:
        ax.text(
```

```
            bar.get_x() + bar.get_width() / 2,
            val / 2,
            f'{val:.1f}%',
            ha='center', va='center',
            fontsize=10, color='white', fontweight='bold'
        )

# Labels inside bars — Subscribed
for bar, bottom, val in zip(bars_yes, no_rates, sub_rates):
    if val > 5:
        ax.text(
            bar.get_x() + bar.get_width() / 2,
            bottom + val / 2,
            f'{val:.1f}%',
            ha='center', va='center',
            fontsize=10, color='white', fontweight='bold'
        )

# Annotate total count above each bar
for i, o in enumerate(order):
    total = poutcome_group.loc[o, 'total']
    ax.text(
        i, 103,
        f'n={total:,}',
        ha='center', va='bottom',
        fontsize=8.5, color='#444441'
    )

# Styling
ax.set_title('Previous Outcome vs Subscription — Does Past Behavior Predict Future?',
             fontsize=12, pad=14)
ax.set_ylabel('Percentage (%)')
ax.set_xticks(x)
ax.set_xticklabels(x_labels, fontsize=10)
ax.set_ylim(0, 112)
ax.set_yticks([0, 25, 50, 75, 100])
ax.yaxis.grid(True, linestyle='--', alpha=0.4)
ax.set_axisbelow(True)
ax.spines[['top', 'right']].set_visible(False)

# Legend
legend = [
```

```
    mpatches.Patch(color='#5DCAA5', label='Subscribed'),
    mpatches.Patch(color='#B4B2A9', label='Not subscribed'),
]
ax.legend(handles=legend, fontsize=9, framealpha=0.3, loc='center')

plt.tight_layout()
plt.show()

# Drop temp column
df_encoded.drop(columns='poutcome', inplace=True)
```

```
Poutcome columns found: ['poutcome_nonexistent', 'poutcome_success']

Subscription rate by previous outcome:
             total  subscribed  not_subscribed  sub_rate
poutcome
nonexistent  39815        3746           36069       9.4
success       1373         894             479      65.1
```



Insights from descriptive statistics and data visualizations

Class Imbalance — The Foundation Problem

The dataset is heavily imbalanced with 88.7% "No" (36,548) and only 11.3% "Yes" (4,640). This is the most critical structural issue in the data.

Implication: A naive model predicting "No" every time would achieve 88.7% accuracy making raw accuracy a misleading metric. This is why its required to used SMOTE oversampling and evaluated models using balanced accuracy, precision, recall,

and F1-score instead.

Call Duration — The Strongest Predictor

Clients who subscribed had an average call duration of approx 553 seconds (9.2 mins) versus approx 221 seconds (3.7 mins) for those who did not, a 2.5× difference. The duration bar chart shows subscribed clients dominate the 721s+ bucket, while non-subscribed clients cluster in the 0–240s range.

Implication: Longer calls indicate genuine engagement and interest. However, duration is a post-call feature you only know it after the call ends, making it a leaky predictor for real-time decision making. Banks should use it for retrospective analysis but not rely on it for pre-call targeting.

Timing Matters — Month of Contact

The monthly trend reveals a striking pattern:

- Best months: March (51.5%), September (42%), October (44%), December (48%)
- Worst month: May (6.5%) — despite being the most frequently used campaign month

Implication: The bank is running the majority of its campaigns in May the single worst-performing month. Shifting campaign focus to Q4 (Sep–Dec) and March could dramatically improve conversion rates without increasing costs. This is a direct, actionable business recommendation.

Job Type — Know Your Audience

Subscription rates vary widely by profession:

- Students (31.4%) and Retired (25.1%) are by far the most receptive
- Blue-collar workers (7%) are the least responsive
- Admin, Technician, and Management cluster around 12–13%

Implication: Students and retirees likely have more financial flexibility, lower time pressure, and greater openness to long-term investment products. The bank should prioritize outreach to these segments and reduce effort on blue-collar workers who show consistently low interest.

Previous Outcome — History Repeats Itself

This is one of the most powerful findings:

- Clients with a previous successful subscription: 65% conversion rate
- Clients who previously declined: only 8%
- Clients never contacted before: 12%

Implication: Past behavior is the single strongest behavioral signal. The bank should build a priority calling list of clients who subscribed in previous campaigns, they convert at 5× the rate of cold prospects. Repeatedly calling clients who previously declined wastes resources.

Economic Indicators — External Forces

The correlation chart reveals that macroeconomic variables are negatively correlated with subscription:

- `nr.employed` (r = −0.35): Higher employment numbers reduce subscription likelihood
- `euribor3m` (r = −0.31): Higher interbank rates reduce subscription
- `emp.var.rate` (r = −0.30): Rising employment variation reduces subscription

Implication: People are more willing to lock money into long-term deposits during economic uncertainty when employment is unstable and interest rates are lower. Banks should intensify campaigns during economic downturns when clients seek safer financial instruments.

Correlation Summary — What Actually Drives Subscription

Ranking features by correlation with target y:

```
            Feature          Direction         Strength          Interpretation

            duration          Positive        Strong (0.41)     Longer calls = higher
interest
      poutcome_success      Positive        Strong (0.29)     Prior subscriber = likely
again
```

```
          previous          Positive        Moderate (0.23    More prior contacts =
familiarity
        nr.employed         Negative        Moderate (-0.35)  High employment = less need
         euribor3m          Negative        Moderate (-0.31)  High rates = less
attractive
       emp.var.rate         Negative        Moderate (-0.30)  Economic instability drives
deposits
         campaign           Negative        Weak (-0.15)      Too many calls = irritation
           age              Positive        Weak (0.03)       Minimal direct effect
```

Campaign Frequency — Less Is More

`campaign` (number of contacts in current campaign) has a negative correlation (−0.15) with subscription.

Implication: The more times a client is called in the same campaign, the less likely they are to subscribe. Repeated calling creates contact fatigue. The bank should implement a maximum call threshold likely 2–3 attempts before deprioritizing a prospect.

Key Business Recommendations

Drawing everything together, the bank should:

- Target students and retirees first highest conversion rate segments
- Prioritize Q4 and March for campaign scheduling avoid May
- Build a re-engagement list of previous subscribers, 65% conversion rate
- Cap call attempts at 2–3 per prospect to avoid contact fatigue
- Monitor economic indicators launch campaigns when euribor3m and nr.employed are low
- Focus on call quality over quantity longer, more engaging conversations drive subscriptions
- Use Random Forest on PCA + SMOTE data your best model at ~82.06% balanced accuracy

In [ ]: